

Abstract

In heterogeneous landscapes, mobile organisms can track habitat conditions across multiple spatial and temporal scales, but selecting one condition often comes at the expense of another. In aquatic ecosystems, trade-offs between temperature and dissolved oxygen (DO) are ubiquitous, and diel cycles of solar heating and photosynthesis can make them dynamic. Heat stress and hypoxia often occur asynchronously because temperature maximums occur in the evening and oxygen minimums occur around dawn. A key question is whether animal behavior can soften temperature-oxygen trade-offs by exploiting this asynchrony. We tested how fish respond to diel variation in DO and temperature within vertically stratified thermal refuges. We first monitored heterogeneity in habitat conditions to quantify diel variation in temperature-DO trade-offs. We then simulated fish depth selection under three strategies: minimizing temperature, maximizing DO, or balancing both variables. Simulations used observed temperature and DO to model depth selection. We then evaluated fish behavior by tagging cutthroat trout with temperature loggers and monitoring their vertical habitat use within enclosures instrumented with temperature-DO sensors. Simulations showed that temperature-minimizing or oxygen-maximizing strategies resulted in constant depth selection, whereas fish that balanced temperature and oxygen exhibited diel vertical movement. Most fish exhibited diel vertical movement that matched simulation, however the movement extent was smaller than expected as fish endured hypoxic conditions in exchange for colder temperatures. We show that diel vertical movement exploits temperature and oxygen asynchrony to soften the trade-off between these variables. These findings likely apply to other aquatic systems where asynchronous stressors constrain habitat use.

Keywords

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Diel vertical movement, temperature-oxygen squeeze, trade-off, floodplain, thermal refuge, habitat compression, asynchrony

Introduction

Environmental heterogeneity can decouple local habitat conditions from regional trends, creating asynchronous variation among habitat patches (Dobrowski, 2011; Fey et al., 2019). While spatial heterogeneity in resources underpins foundational concepts in ecology such as optimal foraging, competition, and niche theory, resource availability can vary both spatially and temporally (Abrahms et al., 2021). This added dimension increases complexity because animals must make decisions under conditions that are constantly changing and potentially conflicting. For example, “Green wave surfing” or the behavior of following the leading edge of vegetation emergence has been observed among many terrestrial herbivores and allows animals to track more favorable conditions across the landscape as they become seasonally available (Van der Graaf et al., 2006). While there is much interest in how animals behaviorally respond to spatio-temporal heterogeneity in single resources at large spatial scales (Armstrong et al., 2016; Abrahms et al., 2021), less is known about responses to multiple resources at fine spatial scales.

In aquatic ecosystems, both temperature and dissolved oxygen (DO) are critically important, because most aquatic organisms are poikilotherms that cannot regulate their body temperature, and because oxygen is scarcer and more variable in water than in air. Higher temperatures both reduce oxygen solubility in water and increase organismal oxygen demand, so we should expect that as temperatures exceed physiological optimums, behavioral responses should be shaped by both temperature and DO (Stehfest et al., 2017). Aquatic habitats ranging from pelagic seas to ponds can exhibit vertical stratification of temperature and oxygen (also referred to as vertical fronts; Logan et al., 2023), with striking variation between adjacent layers

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of water. In some instances, optimal temperatures and sufficient DO can be found in the same depth zone. For example, warm-adapted fishes in tropical marine environments encounter both optimal temperatures and sufficient DO levels in shallow waters, allowing them to fully meet their physiological requirements in a single habitat layer, while deeper habitat is suboptimally cold and hypoxic (Brill et al., 1999; Prince & Goodyear, 2006; Logan et al., 2023). In contrast, cold-adapted taxa often face a trade-off: shallow water offers adequate oxygen but suboptimally warm temperatures, while deeper water is cooler but hypoxic. This vertical compression of suitable habitat is commonly called the temperature-oxygen squeeze (Coutant, 1985; Barrett et al., 2024), and is intensified by climate warming and eutrophication as they compress suitable habitat from opposite ends (Arend et al., 2011; Jane et al., 2024). Several studies have documented fish distributions contracting under temperature-oxygen compression (Coutant, 1990; Lauchlan & Nagelkerken, 2020; Stahl et al., 2025), and, conversely, an experiment that oxygenated a lake hypolimnion expanded the depth range fish occupied (Aku et al., 1997). These effects have been shown in both cold- and cool-water fish guilds (Coutant, 1985; Ebersole et al., 2003), and are defined by the physiological tolerances of fish relative to environmental conditions rather than by absolute values, so they could arise across a broad range of species and systems.

While most prior work has documented vertical temperature-oxygen trade-offs in oceans, lakes, and reservoirs, they also occur in rivers. The main channels of rivers are typically vertically mixed, but floodplains often stratify and can express large gradients in temperature and DO across small spatial extents (Ford et al., 2002; Valett et al., 2005). There is much interest in the potential for floodplain heterogeneity to provide thermal refuge for cold-water fishes (Fernald et al., 2006), but recent work shows that temperature-oxygen trade-offs can constrain

that potential: hypoxia in the bottom layer of stratified backwaters, combined with warm temperatures in the surface layer, confined cold-water fish to $\leq 20\%$ of the water column (Barrett et al., 2024). How behavioral responses might alleviate this trade-off, however, remains poorly understood. Prior work has shown that fish can alleviate trade-offs through foray behavior, i.e., brief excursions into physiologically suboptimal habitat (Rahel & Nutzman, 1994; Roberts et al., 2012; Armstrong et al., 2013). Forays can resolve trade-offs between foraging and physiological suitability because a meal can be acquired quickly, but they should not resolve temperature–oxygen trade-offs, because oxygen uptake and thermal balance are continuous demands that cannot be met during brief excursions.

Animals can also soften trade-offs by tracking temporal variation in conditions, timing their movements to occupy favorable habitat when the optimum shifts. Prior work has mostly considered the temperature-oxygen trade-off to be temporally static. This assumption holds in large water bodies, where thermal inertia buffers temperature against the day–night cycle, and in unproductive systems, where low ecosystem metabolism drives only weak diel fluctuations in DO. However, in smaller and more productive systems, diel variation in both temperature and DO becomes pronounced (Caissie, 2006; Andersen et al., 2017; Maasri et al., 2019). Indeed, there are examples of temperature-oxygen compression with diel variation in a variety of ecosystems, such as the floodplains of large (Barrett et al., 2024) and small (Ebersole et al., 2003) rivers, beaver ponds (Stevenson et al., 2022), shallow lakes (Whalen et al., 2008), and estuaries (Mahardja et al., 2022). Low temperature and low DO occur at night, whereas high temperature and high DO occur during the day, creating asynchronous threats of hypoxia and heat stress for fish.

To increase fitness under diel variation in environmental variables, aquatic organisms often exhibit diel vertical movement (DVM). DVM is among the most widespread movement behaviors in aquatic systems, occurring in all sizes of water bodies and across multiple trophic levels. There are three primary hypotheses for the ultimate causes of DVM: (1) predator avoidance (Zaret & Suffern, 1976), (2) bioenergetic efficiency (Svetlichny & Hubareva, 2005), and (3) feeding opportunity (Hidalgo et al., 2005; Mehner, 2012a), none of which explicitly consider DO. Theory on DVM instead considers DO as a static abiotic gradient that can play a secondary role by constraining the expression of DVM caused by the three primary hypotheses. For example, studies have explored the level of hypoxia that insects (Stratton & Kesler, 2006), zooplankton (Doubek et al., 2018), and (to a lesser extent) fish (Vanderploeg et al., 2009) will incur while exhibiting predator avoidance DVM (i.e., moving deeper during the daytime to avoid light-mediated predation risk). Further, work has explored how DVM into hypoxic waters associates with physiological tolerances (Weisser et al., 2018; Tremblay et al., 2020). However, DO has never been considered as an ultimate cause of DVM, because prior work has emphasized spatial, but not temporal, heterogeneity in DO, and where temporal variation was considered, it was not at the diel scale (Mehner, 2012a; Bandara et al., 2021). While prior work has explored the implications of diel variation in temperature and DO for benthic invertebrate communities, behavioral responses more broadly—and those of vertebrates such as fish in particular—remain poorly understood. Here we use simulation to explore whether diel variation in hypoxia and thermal stress could be an ultimate factor for DVM and we empirically test whether cold-water fish experiencing temperature-oxygen habitat compression exhibit DVM.

To explore how behavioral responses can mediate the strength of dynamic temperature-oxygen trade-offs we: (1) monitored spatio-temporal heterogeneity in temperature and oxygen to

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characterize diel and vertical heterogeneity, (2) simulated fish habitat selection under alternative tactics – minimize temperature, maximize oxygen, or jointly balance temperature and oxygen – to identify whether DVM was an emergent behavior, and (3) recorded fish depth selection *in situ* by tagging fish with high-resolution temperature loggers and releasing them into large net pens to empirically test hypothesized behavioral responses. We hypothesized that fish would exhibit DVM to exploit asynchronous variation in temperature and oxygen constraints. By integrating simulations with empirical observations, we demonstrate that dynamic temperature-oxygen trade-offs are an understudied ultimate driver of diel vertical movement that could shape behavior in a diversity of aquatic ecosystems.

Methods

Study location

The Willamette River basin lies between the Cascade and Coastal Mountain ranges of Western Oregon. Historically, the Upper Willamette River (between Eugene and Salem) was a laterally dynamic and meandering gravel-bed river with side channels, islands, and gravel beds (Wallick et al., 2013; Flitcroft et al., 2023). While these features still exist, they have been significantly reduced since Euro-American settlement (Baker et al., 2004; Gregory et al., 2019). This portion of the river also contains numerous alcoves, i.e., floodplain features with only a downstream connection to the mainstem river. A subset of these waterbodies function as cold-water refuges, providing temperatures up to 10°C cooler than the adjacent mainstem river due to a combination of hyporheic and surface water heat transfer processes (Arrigoni et al., 2008; Gombert et al., 2022). These cold-water habitats are seasonally exploited by cold-water fishes, particularly coastal cutthroat trout (*Oncorhynchus clarkii clarkii*), for which counts peak in July and August (Barrett & Armstrong, 2025). During summer, the mainstem Willamette River

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remains well mixed and normoxic (USGS station 14211720, Willamette River at Portland, OR), with temperatures often exceeding 20°C during peak summer (Barrett & Armstrong, 2025). Cold-water alcoves are often characterized by pronounced vertical stratification and diel variation of DO (Zarnetske et al., 2011; Andersen et al., 2017; Gombert et al., 2022). Surface water temperatures also fluctuate on a diel cycle in time-lagged synchrony with DO saturation (i.e., oxygen saturation is highest during midday, whereas surface water temperatures reach their maximum during early evening). The fluctuating conditions found in these alcoves likely cause the zone of optimal habitat to shift over the 24-hour cycle. While temperature and oxygen fluctuate in rough synchrony numerically (i.e., both reach maximums during the day), they fluctuate asynchronously when considered through the lens of cold-water fish physiology, for which high temperature and/or low DO are limiting.

Physiology of study species

In coastal cutthroat trout from Willamette River tributaries, the critical thermal maximum was 27°C, aerobic scope peaked near 20°C, and metabolic costs rose exponentially with temperature (Anlauf-Dunn et al., 2022). The hypoxia tolerance of coastal cutthroat trout has not been quantified, but studies on closely related rainbow trout (*O. mykiss*) show that growth begins to decline at DO concentrations as high as 7 mg L⁻¹ (Waldrop et al., 2018). A variety of performance metrics have been shown to decline from ~8 mg L⁻¹ to ~4 mg L⁻¹ (Magnoni et al., 2018; Aksakal & Ekinici, 2021) which is reflected in water quality protective standards for cold-water fish, with minimum values ranging from 4 mg L⁻¹ to 6 mg L⁻¹ depending on the time of exposure. Work on other *O. clarki* subspecies found a lethal threshold of ~2 mg L⁻¹ (Wagner et al., 2001).

Spatiotemporal patterning of temperature and oxygen

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In 2020, we collected data on temperature and DO within six known cold-water alcoves on the Upper Willamette River to characterize spatiotemporal variation in DO and temperature, and the resulting trade-offs to fish. These measurements were used alongside data collected in 2021 to quantify temperature–oxygen relationships within cold-water alcove habitats. During the summer, we instrumented each of these alcoves with two separate temperature-DO arrays deployed longitudinally (i.e. along the length of the alcove) near its mid-point and near its terminus (the closed end of the alcove, farthest from the river). Each array was comprised of three temperature-DO loggers (Onset HOBO® U26-001 Dissolved Oxygen Data Logger; DO accuracy to ± 0.2 mg L⁻¹, and temperature to $\pm 0.2^\circ\text{C}$) which were calibrated using 100% water saturated air (Onset Computer Corporation 2016). Loggers were set to record at 10-minute intervals at the surface, mid-, and bottom of the water column. Arrays were deployed in each alcove for a total of 3–4 days.

We then focused our study on a subset of these alcoves for more intensive monitoring to explore both simulated and observed fish behavior. We continuously monitored DO and temperature at two alcoves (Blue Ruin and Norwood) for ~3 weeks during peak summer 2021 (July 25–August 14) to capture more spatially continuous data. In Blue Ruin alcove, we deployed five vertical logger arrays laterally along the deepest portions of the alcove to characterize spatiotemporal heterogeneity in conditions (Figure S1). Each array was instrumented with at least two temperature-DO loggers positioned in the epilimnion (the warm, upper layer of a stratified water column) and hypolimnion (the cold, bottom layer of a stratified water column). We use these terms (epilimnion and hypolimnion) for ease of communication and to describe analogous thermal and oxygen gradients within the study alcoves; however, we acknowledge that the physical processes generating these layers may differ from those that

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produce epilimnetic and hypolimnetic conditions in lakes. Alcove morphology varied across sites, with a mean maximum depth of 2.65 ± 0.7 m (mean $\pm$ SD) across all six alcoves.

Maximum depths reached approximately 4 m at Blue Ruin and 2 m at Norwood. One of the arrays in Blue Ruin (the site-4 array), as well as the array deployed in Norwood alcove, were associated with our fish enclosure study, which is described in greater detail below. These arrays were more densely instrumented to obtain profile data at a finer resolution. Loggers on all arrays were set to record at 5-minute intervals.

Quantifying in situ movement

We sought to quantify the depths and associated temperature and oxygen conditions selected by fish in the wild. While pressure sensors on tags can estimate depth, they lack the accuracy needed for detecting sub-meter changes. Instead, we tagged fish with temperature loggers and inferred depth and oxygen from vertical profiles, following the approach of Barrett et al. (2024).

In the summer of 2021, we externally tagged 31 cutthroat trout with waterproofed Thermocron high-resolution iButtons (DS1922L-F5#, temperature accuracy $\pm 0.5^\circ\text{C}$). These iButtons were modified to reduce weight following methods outlined by Lovegrove (2009) and set to record temperature at 12-minute intervals to maximize data resolution while ensuring adequate battery life and data storage capacity. Each modified iButton was mounted to a dark green 2.5-cm Peterson disc using marine grade epoxy (JB MarineWeld™ syringe two-part adhesive). The Peterson discs had two small gauge holes drilled 3 mm apart and 3 mm in from the disk's outer margin, which were used to suture the mounted iButton to each fish through their anterior dorsal musculature, with tags secured flush to the body. Tag weights ranged from 3.4 g to 5.3 g ($4.1 \text{ g} \pm 0.4$; mean $\pm$ SD) and did not exceed 3.8% of fish body mass (mean mass of

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tagged fish was 274.8 ± 85.0 g; empirical studies have shown tag burdens $\leq 6.7\%$ do not negatively impact fish; [Brown et al., 2010](#)).

Cutthroat trout were captured via electrofishing on July 22 and 23, 2021. Anesthetization, handling, and recovery procedures followed [Liedtke et al. \(2012\)](#). Tags were attached through the anterior dorsal musculature of each fish approximately 5 mm ventral and 5 mm posterior of the first fin ray using a PDS® Plus 4-0 violet-dyed suture with a ½ circle 26-mm needle. After full recovery, [\(i.e., fish had regained equilibrium and normal swimming behavior\)](#), the fish were transferred to net pens. We deployed three net pen enclosures, two in Blue Ruin alcove ([depth ~1.7 m](#)) and one in Norwood alcove ([depth ~1.6 m](#)). Net pens were [made](#) of plastic mesh (1.27 cm x 1.27 cm) formed into a 1.7 m diameter cylinder that extended the full depth of the water column. The bottom of the pen was weighted to the benthos and sewn closed with a panel of mesh. The surface of the pen extended ~0.45 m above the water's surface and was also capped with mesh. Floats were attached around the upper margins of the pen, creating a semi-rigid and secured cylindrical pen that extended the full depth of the water column.

We placed 11 tagged fish in the Norwood alcove enclosure and 20 in the Blue Ruin enclosures (10-fish per enclosure). Stocking densities ranged from 0.8 to 0.87 kg m⁻³ to reduce fish stress and maintain welfare. Research on stocking densities in caged Atlantic salmon (*Salmo salar*) found that fish welfare (determined by blood glucose, cortisol, and body condition) began to decrease at densities above 22 kg m⁻³ [\(Turnbull et al., 2008\)](#). [In total, we collected data from 12-fish in Blue Ruin alcove and 7-fish in Norwood alcove. Ten](#) fish escaped the net pens during the study, and two of the iButtons recovered from Blue Ruin were corrupted.

To characterize temporal shifts in the vertical distribution of temperature and DO [experienced by tagged fish](#), we deployed a vertical array of DO and temperature loggers outside

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and within 1-meter of the net pen. A temperature-DO profile was first taken using a handheld YSI (ProSolo Digital handheld sonde, Yellow Springs, OH, USA; calibrated daily) to determine depth of the thermocline at 12:00 noon. Temperature-DO loggers (MiniDOT® Logger, DO accuracy to $\pm 0.1 \text{ mg L}^{-1}$ and temperature to $\pm 0.1^\circ\text{C}$, or Onset HOBO® U26-001 DO Data Logger, DO accuracy to $\pm 0.2 \text{ mg L}^{-1}$, and temperature to $\pm 0.2^\circ\text{C}$) were set to read at 25 cm below the surface, at the thermocline, and at the hypolimnion. Additional temperature loggers (HOBO Water Temperature Pro v2 Data Logger, temperature accuracy to $\pm 0.2^\circ\text{C}$) were affixed to the line between the bottom and mid-DO logger, between the mid and surface DO logger, and at the water's surface. Net pens were cleared of any debris or buildup twice weekly but were otherwise left undisturbed. Logger membranes were also wiped clean during this maintenance. At the end of the study, fish were removed from the pen, temperature loggers were removed, and fish were released back into the alcove.

Simulating fish movements

We used temperature and DO data from the arrays associated with our fish enclosure study to simulate fish depth selection over a ~1-week period (July 29 - August 5) under three behavioral strategies: selection to maximize DO (DO_{max}), selection to minimize temperature (T_{min}), and selection to maximize a water quality index that includes both temperature and DO (TDO_{bal} , described below). We ran two separate simulations – one for Blue Ruin and one for Norwood.

Two of the temperature-DO loggers at the Blue Ruin site-4 net pen array (loggers at 1.55 m and 0.5 m depth) were fouled with algae during the study and frequently recorded excessively high and low DO values, despite weekly cleaning of the sensors. The nearby site-5 logger array (within 10 m of site-4; Figure S1) provided comparable vertical profiles of DO and temperature

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(depth- and time-matched DO were within $\leq 0.3 \text{ mg L}^{-1}$), and total depths were similar between sites (within 0.1 m). We therefore removed the DO data from the two fouled loggers at the site-4 net pen array and substituted DO data from site-5 at the closest corresponding depths (1.35 m and 0.4 m). This revised dataset was used for the Blue Ruin simulation, and to quantify DO selection by fish from the enclosure study at that site.

Temperature and DO were interpolated between logger depths to generate profiles at 0.1 m vertical resolution across the full water column. Interpolation was performed using a modified linear approach (Barrett et al., 2024), providing a continuous estimate of habitat conditions across depths. To evaluate behavioral strategies, we constructed habitat suitability indices corresponding to $T_{\min}$ and $DO_{\max}$ following Sullivan et al. (2003). Temperature and DO were linearly rescaled to unitless values ranging from 0 (least favorable) to 1 (most favorable). Rescaling was applied across the full temporal window (29 July–5 August), within each site, to ensure a consistent suitability scale and prevent artificial shifts in relative habitat quality among days. For temperature, the minimum observed value was assigned a suitability of 1 and the maximum a suitability of 0 (11.9°C and 23.5°C, resp., at Blue Ruin; 12.3°C and 23.8°C, resp., at Norwood). For DO, the maximum observed value was assigned a suitability of 1 and the minimum a suitability of 0 (9.4 and 2.2 mg L^{-1} resp., at Blue Ruin; 12.1 and 0.7 mg L^{-1} , resp., at Norwood). Intermediate values were interpolated linearly. These suitability indices were combined into a composite temperature–oxygen water quality index (TDO_{bal}), calculated as the geometric mean:

$$\text{Equation 1: } TDO_{\text{bal}} = (T_{\min} * DO_{\max})^{1/2}$$

(Sullivan et al., 2003), representing a composite measure in which poor conditions in either dimension constrain overall suitability. This function is shown graphically in Figure S2. Fish depth selection under each strategy ($T_{\min}$, $DO_{\max}$, and TDO_{bal}) was simulated using a stochastic sampling framework. For each hour, all available depths were assigned their corresponding suitability values. To represent selective behavior, these values were raised to a selection exponent of 20, which increases contrast among depths and preferentially weights higher-quality habitat. The transformed values were then normalized to sum to one, producing a probability distribution across depths. Using this distribution, depths were randomly sampled with replacement ($n = 100,000$ draws) for each hour and strategy, generating empirical distributions of predicted depth use. From these distributions, we calculated the mean depth selection and interquartile range (25–75%) to summarize typical habitat use while minimizing the influence of extreme values.

To reduce high-frequency variability introduced by stochastic sampling, depth trajectories were smoothed using a centered moving average (rolling mean). A window size of seven time steps was applied, such that each value represents the mean of the focal hour and the three preceding and three subsequent hours (± 3 hours). This approach reduces short-term noise while preserving DVM patterns and avoiding the temporal distortion associated with broader smoothing methods.

Analysis

Quantifying diel changes in temperature and oxygen trade-offs

We combined all array data from the summers of 2020 and 2021, which included measurements from the six alcoves instrumented in 2020. Based on time-series plots of DO, we

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Deleted: We combined all array data from the summers of 2020 and 2021. Based on time series plots of the data, we identified 6:00-8:00 as the best overlapping 2-hour window of minimum DO, and 18:00-20:00 as the best overlapping 2-hour window of maximum DO, across all alcoves. To characterize trade-offs in the relationship between temperature and DO, and how they varied among alcoves and between the windows of oxygen minimum and maximum, we used linear mixed effect models following Barrett et al. (2024). We fit separate models for the DO-maximum and DO-minimum time windows. To account for random variation from sample site, we included alcove location as a random intercept. Year was not included as a random intercept because it did not explain a significant portion of variation and did not improve model fit (based on AIC). We also fit separate linear models to the data collected at our two focal alcoves, Blue Ruin and Norwood. All models were fit using the lme4 package in R (Bates et al. 2014). ¶

identified 0600–0800 h as a consistent 2-hour window when DO reached its daily minimum across alcoves, and 1800–2000 h as the corresponding window of daily maximum DO. For simplicity, we hereafter will refer to the nighttime DO minimum as ‘DO-minimum’ and the daytime DO maximum as ‘DO-maximum’. These windows provided standardized periods for evaluating how temperature–oxygen trade-offs differed when oxygen was most limiting versus most abundant. To quantify these trade-offs, we modeled DO as a function of temperature using linear mixed-effects models. We fit separate models for the DO-minimum and DO-maximum windows. Because observations were clustered within alcoves, we included alcove location as a random intercept. Year was evaluated as a potential random effect but did not explain additional variance or improve model fit (based on AIC), so it was excluded. To illustrate site-specific patterns at the fish-holding sites, we also fit separate linear models to data from our two focal alcoves, Blue Ruin and Norwood. All models were fit using the lme4 package in R (Bates et al., 2014).

Quantifying movement patterns

All iButton data analyses were limited to July 29 – August 5, 2021, beginning at least five days after fish capture to allow recovery from tagging-induced stress responses and normalization of behavior (Jepsen et al., 2001, 2005). Fish with externally mounted iButtons often exhibited temperatures intermediate to those recorded at the net pen–associated logger arrays. We used linear interpolation to estimate temperature and DO at 0.2 m depth intervals for both net pen–associated arrays. We then matched fish iButton temperatures to the closest bounding profile temperatures for each time step and used linear interpolation to gain resolution in our fish-depth estimates (Barrett et al., 2024).

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Deleted: All iButton data analyses were limited to the period from August 1 to August 7, 2021, starting at least five days after fish capture to ensure that cortisol levels and swimming behavior had returned to normal (Jepsen et al. 2005). Fish with externally mounted iButtons often exhibited temperatures intermediate to those recorded at the net pen associated logger arrays. We used linear interpolation to create a continuous estimation of temperature and DO (at 0.2 m increments) for both net pen associated arrays. We then matched fish iButton temperatures to the closest bounding profile temperatures for each time step and used linear interpolation to gain resolution in our fish-depth estimates (Barrett et al. 2024).

To test for diel vertical migration (DVM), we classified each depth observation as either day (09:00–21:00) or night (21:00–09:00). For each fish during each 24-hour period, we identified the deepest depth recorded during the day and the shallowest depth recorded at night. We then calculated the daily amplitude of vertical movement as the difference between these depths. To evaluate whether fish consistently occupied deeper depths during the day, we used Wilcoxon signed-rank tests with a one-sided alternative hypothesis (greater than 0). Because amplitude distributions were non-normal, this nonparametric approach was used separately for each site.

Habitat selection

The cyclic and covarying nature of temperature and DO levels within a 24-hour period means that simply characterizing the conditions that fish use provides an incomplete understanding of how changing conditions are mediated by behavioral adjustments such as diel vertical migration. For example, observing that fish occupied areas in the water column with a temperature of 16°C tells us little about preference until we compare those conditions to what was available at the same time, i.e., selection. We therefore quantified selection by relating the conditions fish used to those available within the water column at the same moment in time. To do this, we used resource selection functions (RSFs) in a ‘used–available’ design as is routinely done in habitat selection studies in terrestrial systems (Manly et al. 2012). Specifically, we used conditional logistic regression (sometimes referred to as matched case–control logistic regression or discrete choice models). For each iButton reading, we compared the temperature and DO at these ‘used’ locations to temperature and DO values in 10 random locations (sampled from a uniform distribution spanning the entire depth of the water column) that were ‘available’ for a fish to use at that moment in time but were not selected. Therefore, the values of our response variable were 1 if the datapoint corresponded to a used location, and 0 if it corresponded to an available location. We restricted our analysis to the two-hour periods of lowest and highest DO in Blue Ruin and Norwood alcoves and used 4-days of data (August 1–4, commencing 5 days after fish were released into net pens).

Temperature and DO were scaled to have mean = 0 and sd = 1 and were the sole predictors in the model because our objective was to assess what temperature and DO fish selected given the available temperature and DO cond...

To evaluate the reliability of inferring fish depth (and associated oxygen exposure) from temperature profiles, we quantified how often fish occupied portions of the water column with sufficient vertical thermal structure to permit unique depth assignment. Vertical thermal structure was quantified using temperature differences (ΔT) between loggers at adjacent depths within each net pen–associated logger array. At each time step, ΔT was calculated between adjacent loggers, and the midpoint between each depth pair was used to represent the vertical location of the thermal gradient. We defined the thermal resolvability of each depth band relative to logger measurement uncertainty. An effective temperature uncertainty ($\sigma_{T,eff}$) was calculated by propagating error from the fish-mounted iButtons ($\sigma_{T,fish} = 0.5$ °C) and array loggers ($\sigma_{T,array} = 0.2$ °C):

$$\text{Equation 2: } \sigma_{T,eff} = \sqrt{\sigma_{T,fish}^2 + \sigma_{T,array}^2}$$

following standard uncertainty propagation methods (Taylor, 2022). Because manufacturer-specified accuracies represent conservative maximum error bounds rather than typical operational error, a more permissive resolvability criterion was adopted. A temperature gradient was considered resolvable when:

$$\text{Equation 3: } |\Delta T| \geq 0.5 * \sigma_{T,eff}$$

This criterion identifies gradients that exceed typical measurement variability rather than requiring exceedance of worst-case error bounds. Under this framework, temperature gradients with $|\Delta T| \geq 0.27$ °C were classified as resolvable. Fish depth observations were then matched to the nearest ΔT layer at corresponding timestamps and sites using a nearest-neighbor approach based on the vertical distance between the estimated fish depth and the midpoint of each ΔT layer. This allowed us to quantify the proportion of fish observations occurring within thermally

resolvable portions of the water column. Fish observations assigned to depth bands that did not meet the thermal resolvability criterion ($|\Delta T| < 0.27\text{ }^{\circ}\text{C}$) were removed prior to analysis. All subsequent analyses, figures, and statistical models therefore represent only observations for which fish depth and associated DO exposure could be resolved with confidence.

To test for DVM we used a linear mixed-effects modeling framework to evaluate whether fish depth varied between periods of minimum and maximum DO. DO conditions were summarized as 4-hour rolling means of the minimum DO available within the water column, and for each day we identified the most hypoxic (DO-minimum) and most oxic (DO-maximum) 4-hour periods. For each fish on each day (fish-day), we calculated the median depth within each 4-hour window. Median fish depth was modeled as a function of DO period (minimum vs. maximum), site (Blue Ruin vs. Norwood), and their interaction. The interaction term tested whether the effect of DO period on fish depth differed between sites. Fish ID was included as a random effect with a random intercept and random slope for DO period, and date was included as a random intercept. Models were fit using restricted maximum likelihood (REML) using the lme4 package in R.

Results

Vertical and diel variation in temperature and DO

Both Blue Ruin and Norwood alcoves exhibited strong thermal stratification and pronounced diel cycling in DO and temperature (Figure 1; Table S1). Surface temperatures varied substantially over the diel cycle, ranging from 17.3–23.4°C in Blue Ruin and 19.9–24.8°C in Norwood, whereas bottom-water temperatures remained comparatively stable at 12.0–13.3°C and 12.4–13.6°C, respectively (Table S1). DO also varied strongly over the

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diel cycle, with the largest fluctuations occurring in the deepest portions of Norwood (Figure 1). Average epilimnetic DO ranged from 3.3–9.0 mg L⁻¹ in Blue Ruin and 6.2–11.1 mg L⁻¹ in Norwood, while hypolimnetic DO ranged from 2.4–4.9 mg L⁻¹ in Blue Ruin and 0.8–8.5 mg L⁻¹ in Norwood (Table S1).

In both alcoves, minimum DO concentrations occurred during the early morning and maximum concentrations occurred during the early evening (Figure 1; Figures S3–S4). The timing of these periods differed slightly between alcoves, with the DO-minimum and maximum occurring approximately from 0500–0700 h and 1700–1900 h in Blue Ruin, and from 0600–0800 h and 1800–2000 h in Norwood. These differences likely reflect variation in alcove orientation and resulting patterns of solar exposure.

We observed a clear trade-off between cool temperature and high DO, as illustrated by a positive linear relationship between the two variables, expressed across six alcoves during both the morning and evening (Figure 2, Table S2, Figure S5). However, the shape of the trade-off differed between the DO-maximum and DO-minimum periods. In all alcoves there was a contracted range of both temperature and DO, meaning fish had reduced potential to trade-off cool temperature for higher DO. At five of the six alcoves, the slope was also more positive during the DO-minimum, meaning that a given level of cooling was accompanied by a larger reduction in DO (i.e., the trade-off was stronger). Notably, we found the opposite result in the Blue Ruin alcove, where the water column, during the DO-minimum, was more homogeneously low in temperature and DO. This meant that fish had limited potential to trade off temperature for oxygen during the period of the DO-minimum.

Diel patterns of fish depth predicted by alternative behavioral responses

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Our simulation yielded three distinct patterns of fish depth use across the diel cycle (Figure 3a & 3c). A fish that maximized DO ($DO_{\max}$) would hold near the water's surface, while a fish that minimized temperature ($T_{\min}$) would hold near the bottom. In contrast to these stationary behaviors, a fish that considered both temperature and oxygen (TDO_{bal}), exhibited diel vertical movements; shifting towards the surface during the late night and early morning when DO at depth became limiting, and moving deeper during the day when surface temperatures were warmer and deeper habitats became more oxygenated. In both the Blue Ruin and Norwood simulations, the $DO_{\max}$ strategy showed a slight midday dip from the surface on several days, likely reflecting elevated DO concentrations at deeper depths within the epilimnion during that time step (approximately 1 mg L^{-1} greater than surface DO). Overall, our simple simulation illustrates how DVM is an emergent property in systems where temperature and DO present spatial trade-offs that vary over the diel cycle (i.e., due to solar absorption, photosynthesis, and respiration).

Fine scale diel vertical movements in captive fish

Tagged individuals consistently occupied thermally structured layers rather than well-mixed regions of the water column, indicating that temperature-based depth inference was reliable for the majority of observations. The proportion of time spent in resolvable thermal structure was 65.3% at Blue Ruin and 71.3% at Norwood, while homogeneous (unresolvable) conditions accounted for 34.7% and 28.7% of observations, respectively. All observations occurring within non-resolvable portions of the water column were excluded from subsequent analyses. The fork length of tagged fish was $286.6 \pm 58.3 \text{ mm}$ (mean $\pm$ SD). Fork length did not differ significantly between fish captured in the two alcoves (Welch's two-sample t-test, $t = -0.57$, $df = 28.73$, $p = 0.575$).

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Cutthroat trout monitored *in situ* did not remain at constant depth but instead exhibited diel vertical movements (Figure 4, Figures S6–S8). Median fish depth varied significantly between DO periods, with individuals occupying deeper depths during periods of maximum DO. At Blue Ruin, fish occupied an average depth of 1.10 m during DO-maximum periods and were 0.19 ± 0.05 m shallower during DO-minimum periods (linear mixed-effects model: $\beta = -0.186$, $SE = 0.047$, $t = -3.92$, $p = 0.001$, $n = 19$ fish, 280 observations). A similar pattern was observed at Norwood, where the estimated depth shift between DO-maximum and DO-minimum periods was approximately 0.23 m. The effect of DO period did not differ significantly between sites (period $\times$ site interaction: $\beta = -0.045$, $SE = 0.080$, $t = -0.56$, $p = 0.58$), indicating no evidence that the magnitude of depth shifts varied between Blue Ruin and Norwood. Random effects indicated substantial among-individual variation in both baseline depth ($SD = 0.15$ m) and responses to DO conditions ($SD = 0.14$ m). Overall, fish occupied deeper depths during DO-maximum periods on 75.3% of fish-days in Blue Ruin (67 of 89 fish-days) and 97.4% of fish-days in Norwood (38 of 39 fish-days).

Fish occupied markedly different oxygen environments during DO-minimum and DO-maximum periods, whereas occupied temperatures differed little between periods. At Blue Ruin, fish occupied water averaging 2.88 ± 0.38 mg L⁻¹ DO during DO-minimum periods and 4.87 ± 0.57 mg L⁻¹ DO during DO-maximum periods; corresponding occupied temperatures were $13.6 \pm 1.3^\circ\text{C}$ and $13.7 \pm 0.8^\circ\text{C}$, respectively. At Norwood, fish occupied water averaging 2.61 ± 0.98 mg L⁻¹ DO during DO-minimum periods and 7.86 ± 0.74 mg L⁻¹ DO during DO-maximum periods, while occupied temperatures averaged $14.5 \pm 1.2^\circ\text{C}$ and $14.6 \pm 1.0^\circ\text{C}$, respectively.

The diel vertical movements expressed by tagged fish broadly resembled the patterns predicted by our simulations (Figure 3). Although individual responses varied (Figure 4), pooled

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depth-use patterns revealed consistent DVM across both sites. Fish occupied shallower habitats at night and deeper habitats during the day, with diel depth shifts of approximately 0.2–1.3 m. A smoothed fit of depth versus time of day, pooled across individuals and days within each site, revealed distinct alcove-specific patterns in vertical movement (Figure 3B and 3D). Fish at both sites occupied shallower habitats at night and deeper habitats during the day, but the timing and magnitude of these depth shifts differed between alcoves. At Blue Ruin, fish changed their depth more gradually and occupied deeper habitats across a broader portion of the daytime period. In contrast, fish at Norwood exhibited a more abrupt transition between shallow and deep habitats, reaching their shallowest depths during the early morning before descending rapidly later in the morning.

Although the simulation predicted that the TDO_{bal} fish would move close to the surface during nighttime hours, tagged fish remained well below the surface throughout the diel cycle (Figure 3B & 3D, Figure S6, S7, & S8). These observed depth trajectories were generally constrained to the metalimnion and upper hypolimnion, suggesting fish actively avoided the warmer, oxygen-rich surface waters. As a result, they experienced thermal and oxygen conditions that fell between those selected by the T_{min} and TDO_{bal} strategies in our simulation.

Environmental conditions experienced by tagged fish

To visualize the thermal and oxygen environments occupied by fish, we overlaid observed fish depth trajectories on diel profiles of temperature and DO (Figure 5; Figures S9–S10). Across both sites, fish remained below the warm surface layer throughout the diel cycle and occupied a relatively narrow depth range. On the representative day (Figure 5), fish exhibited similar diel movement patterns at both alcoves, occupying shallower depths during nighttime and early morning hours, descending to their greatest depths around midday, and

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returning to shallower depths during the evening. In Blue Ruin (Figure 5A–B), fish generally occupied depths between approximately 0.7 and 1.3 m, remaining below the warmest surface waters while avoiding the deepest portion of the water column. In Norwood (Figure 5C–D), fish occupied slightly deeper depths overall but displayed a similar pattern of daytime descent and nighttime ascent. These diel vertical movement patterns were highly repeatable across the 8-day study period (Figure S9 – S10), with fish navigating trade-offs between increasing temperatures near the surface and reduced oxygen availability at depth.

Discussion

Spatial and temporal heterogeneity in habitat conditions can present animals with trade-offs that shift over time. A key challenge is to understand whether animal behavior can soften these constraints by exploiting asynchronous temporal variation. Here, we quantified diel variation in spatial relationships between temperature and DO in floodplain habitats that provide thermal refuge for cold-water fish. Along a depth gradient, cooler temperatures came at the expense of lower DO, but the shape of this relationship varied over the diel cycle because the two stressors were temporally offset. Simulating behavioral responses to these observed patterns showed that fish responding to a single variable selected a constant depth, whereas fish balancing temperature-DO trade-offs exhibited DVM (Figure 4a & 4c). We empirically corroborated this hypothesis in wild cutthroat trout held *in situ*. The majority of fish in our study exhibited fine-scale DVM, moving deeper during the day and shallower at night (Figure 4b, 4d, Figure 5). This pattern enabled fish to avoid the lowest oxygen levels expressed while selecting for cool temperatures that reduced metabolic costs (Anlauf-Dunn et al., 2022). Surprisingly, fish did incur substantial hypoxia by not moving higher in the water column at night, which likely reflects a cost-minimizing strategy that entails low oxygen demand and high energetic efficiency (Barrett

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& Armstrong, 2025). Taken together, our results show that diel variation, and adaptive behavior (i.e., DVM) can soften the constraints of DO on thermal refuge use, also known as the ‘temperature-oxygen squeeze’. Temperature-DO trade-offs are common in freshwater ecosystems, and our results are supported by first principles, so they are likely relevant to other aquatic ecosystems where similar asynchronous stressors occur, e.g., estuaries, wetlands, ponds, shallow lakes, and floodplain rivers.

Our simulation of fish habitat selection found that fish held at a relatively constant depth when responding to temperature or DO in isolation, but switched to DVM when balancing both variables simultaneously (Figure 4a & 4c). This suggests a novel mechanism for DVM that does not rely on predation risk or foraging, but instead emerges from dynamic temperature–oxygen trade-offs. Testing this hypothesis in the wild on a fish population known to experience such trade-offs (Barrett et al., 2024), we observed analogous patterns of DVM with similar timing, albeit at a smaller amplitude than predicted by the simulations (Figure 4b & 4d). We did not expect the simulations to precisely predict fish depths, as our composite water quality index is a coarse proxy for habitat suitability rather than a mechanistic representation of performance; we favored this approach because, although thermal performance is relatively well characterized, the physiological costs of short-term hypoxia and its interaction with temperature remain poorly quantified and likely vary among individuals (Faust et al., 2004; Metcalfe et al., 2016). Indeed, our mixed-effects model revealed substantial variation among fish in both baseline depth use and the magnitude of depth shifts associated with oxygen conditions. Some of the observed variation likely reflects differences among individuals and local environmental conditions. For example, the diel amplitude of hypolimnetic DO was much greater in Norwood alcove, while epilimnetic DO remained relatively high throughout the diel cycle (Table S1, Figure S3 & S4). However,

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surface temperatures in Norwood remained elevated even at night, thereby shifting diel movement deeper in the water column. In contrast, nighttime temperatures in the epilimnion of Blue Ruin were considerably lower and DO remained adequate, providing more favorable conditions for upward movement. These differences in thermal and oxygen regimes may have contributed to specific patterns in vertical habitat use observed among tagged fish. Nevertheless, the population-level pattern was clear: fish consistently exhibited DVM as predicted by the TDO_{bal} simulation. The simulation was intended as a relative framework describing responses to temperature–DO trade-offs rather than an attempt to define species-specific physiological optima. Because environmental conditions at our study sites spanned a broad range from stressful to favorable for cutthroat trout, this approach resolved meaningful differences in habitat quality.

The fish in our study appear to be relatively tolerant of temporary exposure to low DO.

Though they avoided the lowest DO levels in the water column, they could have experienced higher DO minimums by moving farther above the thermocline. For example, the US Environmental Protection Agency criteria for absolute minimum DO levels for cold-water fish is 4 mg L^{-1} (USEPA, 1987), and the cutthroat in our study could have stayed above this level by selecting $\sim 3 - 4^{\circ}\text{C}$ warmer water, but they did not. Matthews and Berg (1997) found similar patterns where rainbow trout aggregated near groundwater seeps, which provided cooler water with lower DO in stratified pools. Despite the availability of warmer ($22 - 24^{\circ}\text{C}$) surface waters with higher DO, the trout selected cooler habitat with DO as low as $\sim 1.5 \text{ mg L}^{-1}$. We suggest that the reason fish incur hypoxia in exchange for cooler temperatures is because they have low oxygen demand. Our prior work (Barrett & Armstrong, 2025) found that when cutthroat trout move to floodplain refuges, they reduce their respiration rates by 65-70% through a combination

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of thermoregulation and reduced activity. This has two implications for conservation: (1) thermal refuges may still function to support fish even if they fail to meet DO guidelines and (2) temperature and fish behavior (i.e. foraging and activity) likely mediate whether hypoxia limits the function of refuges.

The observation that fish incurred hypoxic conditions to access water below 15°C could suggest that our focal populations have especially low thermal tolerance. However, recent work on our focal species found that temperatures up to 19°C did not decrease aerobic scope (Anlauf-Dunn et al., 2022), suggesting that fish could have occupied warmer, better-oxygenated habitats without exceeding physiological optima for aerobic performance or growth. We therefore suggest that cumulative energetic consequences, rather than avoidance of acute thermal stress, drove habitat selection. While sampling fish in cold-water alcoves as part of a separate study, we observed high densities of fish with small ration sizes (Barrett & Armstrong, 2025). Cooler temperatures reduce metabolic costs and can extend the duration over which fish can survive without foraging (Fry, 1947). Thus, when fish balance temperature–DO trade-offs in thermal refuges, where negative energy budgets are common (Hahlbeck et al., 2022), incurring some level of hypoxia in exchange for lower metabolic costs may reduce the risk of starvation during periods of thermal stress.

The majority of tagged individuals exhibited patterns consistent with DVM (Figure 4), though some individuals showed weaker or less consistent vertical shifts (Figure S6 & S7). Though there was some intra- and inter-individual variation in expression, the central tendency across individuals was consistent with a vertically cycling movement pattern constrained to ~20% of the water column, ranging between the metalimnion and upper hypolimnion (Figure 5,

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Figure S6-S8). This depth range also corresponds to the portion of the water column exhibiting the strongest vertical gradients in temperature and DO.

Behavioral variation in DVM could reflect variation in genetics (Alonzo, 2015), individual thermal performance traits (Anlauf-Dunn et al., 2022) or energy reserves (Moran et al., 2021). Fish ration size affects the optimal temperature for growth and could influence habitat selection by fish (Mac, 1985; Sogard & Olla, 1996). While holding fish in captivity can reduce feeding levels, the fish in our study feed minimally while in thermal refuges (Barrett & Armstrong, 2025), so any changes in their energy intake was likely small and would not substantially alter thermal habitat selection. Mehner's (2012) review of fish DVM argued that many factors can influence its expression, and that multiple drivers—such as bioenergetic efficiency and predation risk—may interact to make observed patterns profitable. We acknowledge that our study cannot definitively rule out alternative explanations for the DVM we observed. However, responses to temperature and oxygen remain the most compelling explanation. We did not observe DVM and then attempt to explain it post hoc; rather, we documented fish balancing temperature–oxygen trade-offs in their daytime depth selection (Barrett et al., 2024) and hypothesized *a priori* that they would exhibit DVM in response to diel changes in the shape of this trade-off. In this prior daytime study, cutthroat trout in cold-water alcoves selected intermediate combinations of temperature and DO that occurred in a narrow band of depths near the thermocline. Importantly, fish depth tracked among-site variation in thermocline depth, a pattern that would not be expected if factors such as light-mediated predation risk were the primary drivers of depth selection. A foraging-related hypothesis for DVM also seems unlikely because, as discussed previously, the fish in our study occupied thermal refuge habitats that provide minimal feeding opportunities. The most plausible

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alternative explanation is that fish tracked a single optimum rather than balancing temperature and oxygen. While such a mechanism could also generate diel movement given the spatiotemporal variation in our system, this was not supported by our data: fish occupied a broad range of temperatures and displayed high individual and intra-individual variation in temperature selection (Figure S6 & S7).

Horizontal movements between thermal refuges and adjacent mainstem habitats represent another potential strategy for coping with environmental stressors. For example, juvenile steelhead in the Klamath River have been documented making repeated movements between thermal refuges and warmer mainstem habitats, balancing the thermal benefits of refuge use against foraging opportunities in the adjacent mainstem (Brewitt & Danner, 2014). However, multiple lines of evidence suggest that such movements are uncommon among cutthroat trout occupying cold-water alcoves in our study system. Fish in cold-water alcoves implanted with temperature-sensing iButtons rarely exhibited body temperatures consistent with repeated excursions into warmer mainstem habitats (Barrett et al., 2024). Similarly, diets of alcove fish were dominated by alcove-associated prey taxa and rarely included taxa characteristic of mainstem habitats (i.e., EPT taxa; Barrett & Armstrong 2025). These observations suggest that fish likely remain within cold-water alcoves during peak summer and use diel movements to balance temperature–DO trade-offs rather than making repeated forays into the mainstem. We acknowledge that fish may also exploit fine-scale three-dimensional heterogeneity in temperature and DO within alcoves. However, as in most studies of DVM, our analyses were limited to vertical habitat selection because resolving dynamic 3-D temperature–oxygen landscapes at scales relevant to fish behavior is not currently feasible in our system.

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There is a growing body of work demonstrating habitat compression due to temperature-oxygen constraints in lakes and reservoirs (Coutant 1985; Swales 2006; McLaren et al. 2023), lentic habitats on a river floodplain (Barrett et al. 2024), and in some marine habitats (Prince and Goodyear 2006; Deutsch et al. 2023). By combining simulation, fine-scale environmental monitoring, and behavioral telemetry, we showed that this habitat compression is dynamic on diel timescales and generates DVM that softens trade-offs. Future research could employ a similar methodology to quantify diel variation in habitat compression and the extent to which it drives DVM. Because the behavior we documented was predicted by first principles, our findings are likely to be broadly relevant across aquatic systems where diel variation in oxygen and temperature stressors co-occur. ¶

There is a growing body of work demonstrating habitat compression due to temperature-oxygen constraints in lakes and reservoirs (Coutant, 1985b; Swales, 2006; McLaren et al., 2023), lentic habitats on a river floodplain (Barrett et al. 2024), and in some marine habitats (Prince & Goodyear, 2006; Deutsch et al., 2023). By combining simulation, fine-scale environmental monitoring, and behavioral telemetry, we showed that this habitat compression can be dynamic on diel timescales, generating DVM that softens trade-offs. This result is particularly relevant to river restoration, where efforts to restore natural processes (Beechie et al., 2010) often generate patches of lentic habitat (e.g., floodplain backwaters and beaver ponds) with dynamic temperature-oxygen trade-offs. Because fish can behaviorally exploit these diel dynamics, such habitats may support species better than static assessments of temperature and oxygen would predict. Future research could employ a similar methodology to test for this behavior in other water bodies, or use new experimental methods for stratifying temperature and oxygen in the laboratory (Bodamer & Bridgeman, 2014) to better resolve mechanisms of habitat selection. Because the behavior we documented was predicted by first principles, our findings are likely to be broadly relevant across aquatic systems where diel variation in oxygen and temperature stressors co-occur.

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